Chapter 8, Lecture 5

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Multinomial distribution

Recall the multinomial distribution, presented in the textbook at pages 73-74, and 272.

It is a discrete distribution and an extension of the Binomial distribution. We have n independent trials. Each trial has m possible outcomes, of type $1, \ldots, m$. We observe type k with probability p_k. For $k = 1, \ldots, m$, X_k counts the number of trials (out of n) with outcome of type k. Since $\sum_{i=1}^m X_m$ is the total number of trials, it is equal to n. The likelihood function in this case is therefore

$$\begin{aligned} \text{lik}(p_1, \ldots, p_m) &= \mathbb{P}\{X_1 = x_1, \ldots, X_m = x_m\} \\ &= \frac{n!}{x_1! \cdots x_m!} p_1^{x_1} \cdots p_m^{x_m} \end{aligned}$$

provided $\sum_{i=1}^m x_i = n$, $p_k \geq 0$ and $\sum_{k=1}^m p_k = 1$.
You will quickly find out that maximizing this function yields nonsense estimates $(+\infty)$. However, we have to ensure that the condition $\sum_{k=1}^m p_k = 1$ holds, that is, we need to maximize over a restricted parameter space. To guarantee this linear restriction, we use the method of Lagrange multipliers. Let

$$\begin{aligned} \ell(p_1, \ldots, p_m) &= \log \binom{n}{X_1, \ldots, X_m} + \sum_{k=1}^m X_i \log p_k \\ G(p_1, \ldots, p_m, \lambda) &:= \ell(p_1, \ldots, p_m) + \lambda(\sum_{k=1}^m p_k - 1) \end{aligned}$$

Since

$$\frac{\partial G(p_1, \ldots, p_m, \lambda)}{\partial \lambda} = \sum_{k=1}^m p_k - 1$$

this gives that $\widehat{p}_1, \ldots, \widehat{p}_m$ satisfy the linear constraint. Now, taking the partial derivative with respect to p_i yields

$$\frac{\partial G(p_1, \ldots, p_m, \lambda)}{\partial p_i} = \frac{X_i}{p_i} + \lambda$$

for each $i = 1, \ldots, m$. This implies

$$X_k = -\widehat{\lambda}\widehat{p}_k$$

and summing over k yields

$$\sum_{k=1}^{m} X_i \;=\; -\widehat{\lambda}\sum_{k=1}^{m}\widehat{p}_k \iff n \;=\; -\widehat{\lambda}$$

so that we arrive at the estimates

$$\widehat{p}_k = \frac{X_k}{n}$$

which is the cell count over the total count.

The properties of these estimates follow from the fact that X_k has a Binomial(n, p_k) distribution with mean np_k and variance $np_k(1-p_k)$. Hence $\widehat{p}_k$ is asymptotically normal with

$$\mathbb{E}[\widehat{p}_k] = p_k \;\text{ and }\; \mathrm{Var}(X_k) = \frac{p_k(1-p_k)}{n}.$$

Special case: Hardy-Weinberg

The Hardy-Weinberg law states that if gene frequencies are in equilibrium, genotypes AA, Aa and aa occur in a population according to frequencies

$$p_1 = (1-\theta)^2,\; p_2 = 2\theta(1-\theta),\; p_3 = \theta^2$$

for some parameter $\theta \in [0, 1]$. See page 273 in your textbook for details.

Note that $m = 3$, $p_k = p_k(\theta) \in [0, 1]$ and the restriction $p_1 + p_2 + p_3$ automatically holds for all $\theta \in [0, 1]$. In the population of Hong-Kong (1937), we found $x_1 = 342$, $x_2 = 500$ and $x_3 = 187$ from a total of $n = 1,029$ cases.

Let us find the MLE using the standard recipe.

$$\begin{aligned}
\text{lik}(\theta) &= \frac{n!}{X_1!X_2!X_3!}(1-\theta)^{2X_1}\{2\theta(1-\theta)\}^{X_2}\theta^{2X_3} \\
\ell(\theta) &= \log\frac{n!}{X_1!X_2!X_3!} + 2X_1\log(1-\theta) + X_2\log(2\theta(1-\theta)) + X_3\log(\theta^2) \\
&= \log\frac{n!}{X_1!X_2!X_3!} + (2X_1+X_2)\log(1-\theta) + (X_2+2X_3)\log(\theta) + X_2\log(2) \\
\ell'(\theta) &= -\frac{2X_1+X_2}{1-\theta} + \frac{X_2+2X_3}{\theta}
\end{aligned}$$

This gives

$$\begin{aligned}
\ell'(\theta)=0 &\implies \frac{2X_1+X_2}{1-\theta} = \frac{X_2+2X_3}{\theta} \\
&\implies X_2+2X_3 = \widehat{\theta}(2X_1+X_2+X_2+2X_3) = 2n\widehat{\theta} \\
&\implies \widehat{\theta} = \frac{X_2+2X_3}{2n}
\end{aligned}$$

In the Hong-Kong example, we find

$$\widehat{\theta} = \frac{2\cdot 187+500}{2\cdot 1,029} \approx 0.4247.$$

This choice leads to the HW-model-based MLEs

$$p_1(\widehat{\theta}) \approx 0.3310,\ p_2(\widehat{\theta}) \approx 0.4887,\ p_3(\widehat{\theta}) \approx 0.1804.$$

The non-model based MLEs, $\widehat{p}_i = X_i/n$ yield

$$\widehat{p}_1 \approx 0.3324,\ \widehat{p}_2 \approx 0.4859,\ \widehat{p}_3 \approx 0.1817.$$

Thee are really close. Later we will formally test the *hypothesis* that the HW model holds. This is a simpler model (1 parameter, versus 3 (or actually 2, since if I give you p_1 and p_2, you know p_3 since they sum to one, right?) and it has a biological motivation / interpretation. Alternatively, we can compare the observed values X_i with the expected values under the HW-model $np_i(\theta)$, or rather their estimates $np_i(\widehat{\theta})$: This

X_i	342	500	187
E_i	340.587	502.826	185.587

looks almost too good to be true. We will return to this example several times.

Finally, we derive some properties. First, we compute the expected value

$$\mathbb{E}[\widehat{\theta}] = \frac{1}{2n}\{\mathbb{E}[X_2] + 2\mathbb{E}[X_3]\} = \frac{1}{2}(p_2 + 2p_3) = \frac{1}{2}\{2\theta(1-\theta) + 2\theta^2\} = \theta$$

so our estimate is unbiased. The variance calculation is a bit more involved. First recall that

$$\mathrm{Var}(X_2) = np_2(1-p_2)$$

and

$$\mathrm{Cov}(X_2, X_3) = -np_2p_3.$$

Let's prove the last identity. Define

$$Y_{ki} = 1\{\text{trial } i \text{ has outcome } k\}, \qquad i = 1, \ldots, n,\ k = 2, 3.$$

This means that

$$X_2 = \sum_{i=1}^{n} Y_{2i}, \qquad X_3 = \sum_{i=1}^{n} Y_{3i}$$

and, using the co-linearity of the covariance, we get

$$\begin{aligned}
\mathrm{Cov}(X_2, X_3) &= \sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(Y_{2i}, Y_{3j}) \\
&= \sum_{i=1}^{n} \mathrm{Cov}(Y_{2i}, Y_{3i}) \qquad \text{since the trials are independent} \\
&= \sum_{i=1}^{n} (\mathbb{E}[Y_{2i}Y_{3i}] - \mathbb{E}[Y_{2i}]\mathbb{E}[Y_{3i}]) \\
&= -np_2p_3
\end{aligned}$$

Now we are in the position to calculate the variance of the MLE, as

$$\begin{aligned}
\mathrm{Var}(\widehat{\theta}) &= \frac{1}{4n^2}\left\{\mathrm{var}(X_2) + 4\mathrm{Var}(X_3) + 4\mathrm{Cov}(X_2, X_3)\right\} \\
&= \frac{1}{4n^2}\left\{np_2(1-p_2) + 4np_3(1-p_3) - 4np_2p_3\right\} \\
&= \frac{1}{4n}\left\{p_2(1-p_2) + 4p_3(1-p_3) - 4p_2p_3\right\} \\
&= \frac{1}{4n}\left\{2\theta(1-\theta)(1-2\theta(1-\theta)) + 4\theta^2(1-\theta^2) - 8(1-\theta)\theta\cdot\theta^2\right\} \\
&= \frac{2\theta(1-\theta)}{4n}\left\{1 - 2\theta(1-\theta) + 2\theta(1+\theta) - 4\theta^2\right\} \\
&= \frac{\theta(1-\theta)}{2n}
\end{aligned}$$

Now we can also compute the *estimated standard error* of our estimate

$$\text{s.e.}(\widehat{\theta}) = \sqrt{\frac{\theta(1-\theta)}{2n}} \approx \sqrt{\frac{\widehat{\theta}(1-\widehat{\theta})}{2n}} \approx 0.011.$$